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Orders, Fermat and Euler

Level 4 Olympiad · Number theory

How the order of a number modulo n controls divisibility of powers, with Fermat, Euler, primitive roots and lifting the exponent.

By the end you can: You will be able to find and use the order of a number modulo n, and prove that it divides $\varphi(n)$.; You will be able to pin down the prime factors of expressions such as $a^m + 1$ using orders.; You will be able to find the exact power of a prime dividing $a^n - b^n$ with lifting the exponent..

Lesson

Here is a question that looks as if it needs a computer. Does 7 ever divide $2^n + 1$? Write out the powers of 2 modulo 7: they run $2, 4, 1, 2, 4, 1, \ldots$, repeating every three steps, and $-1 \equiv 6$ never turns up. So the answer is no, for every n, and we have proved it by looking at three numbers. Now try $3^n + 1$. The powers of 3 modulo 7 are $3, 2, 6, 4, 5, 1$ and then repeat, and $3^3 \equiv 6 \equiv -1$, so $7 \mid 3^n + 1$ exactly when $n \equiv 3 \pmod 6$.

Everything in this unit grows out of that observation: powers modulo n go round in a cycle, and the length of the cycle decides which exponents work.

The order of a number

Let $n \geq 2$ and let a be an integer with $\gcd(a, n) = 1$. The **order** of a modulo n, written $\operatorname{ord}_n(a)$, is the smallest positive integer d with $a^d \equiv 1 \pmod n$.

Why does such a d exist? There are only n residues, so two of $a^0, a^1, \ldots, a^n$ agree, say $a^i \equiv a^j$ with $i < j$. Because a is coprime to n we may cancel it i times, which leaves $a^{j-i} \equiv 1$. So some positive power is 1, and hence there is a smallest one.

> **Order Lemma.** Let $d = \operatorname{ord}_n(a)$. Then $a^k \equiv 1 \pmod n$ if and only if $d \mid k$. Consequently $a^i \equiv a^j \pmod n$ if and only if $i \equiv j \pmod d$.

Proof. If $k = dm$ then $a^k = (a^d)^m \equiv 1$. Conversely, suppose $a^k \equiv 1$ and divide: $k = qd + r$ with $0 \leq r < d$. Then $1 \equiv a^k = (a^d)^q a^r \equiv a^r$. Since $r < d$ and d is the smallest positive exponent giving 1, we must have $r = 0$. The second statement follows by cancelling the smaller power. □

Two consequences of the lemma are used constantly. First, if $a^m \equiv 1$ and $a^k \equiv 1$ then d divides both, so $a^{\gcd(m,k)} \equiv 1$. Second, to show that the order is exactly d, it is enough to check that $a^d \equiv 1$ and that $a^{d/q} \not\equiv 1$ for every prime $q \mid d$, because every proper divisor of d divides some d/q.

> **Example 1.** Find the smallest positive integer n such that 1001 divides $2^n - 1$.
>
> *Solution.*
> Multiplying by 2 modulo 1001 until 1 appears works, but it is slow and one slip ruins it. Better to split the modulus: $1001 = 7 \cdot 11 \cdot 13$, and $1001 \mid 2^n - 1$ if and only if each of 7, 11, 13 divides $2^n - 1$. By the Order Lemma that happens if and only if n is a multiple of each of the three orders, so the answer is their lowest common multiple.
> Modulo 7, $2^3 = 8 \equiv 1$ and $2 \not\equiv 1$, so the order is 3. Modulo 11, $2^5 = 32 \equiv -1$, so $2^{10} \equiv 1$ while $2^5 \not\equiv 1$ and $2^2 = 4 \not\equiv 1$; the primes dividing 10 are 2 and 5, so by the test above the order is 10. Modulo 13, $2^6 = 64 \equiv -1$, so $2^{12} \equiv 1$, while $2^6 \not\equiv 1$ and $2^4 = 16 \equiv 3 \not\equiv 1$; the primes dividing

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12 are 2 and 3, so the order is 12.
Hence the answer is $\operatorname{lcm}(3, 10, 12) = 60$.

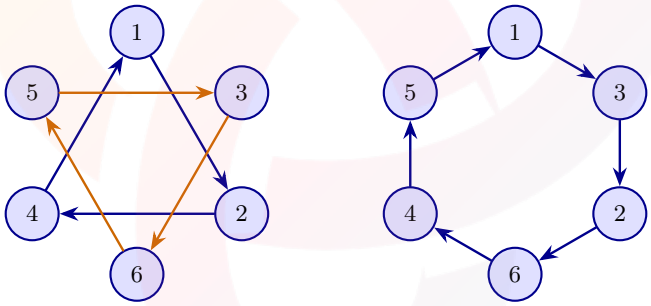
Fermat and Euler

Write $\varphi(n)$ for the number of integers in $\{1, 2, \ldots, n\}$ that are coprime to n. For a prime, $\varphi(p) = p-1$.

Euler's Theorem. If $\gcd(a, n) = 1$ then $a^{\varphi(n)} \equiv 1 \pmod{n}$. In particular (Fermat's Little Theorem) $a^{p-1} \equiv 1 \pmod{p}$ for a prime p not dividing a. Combined with the Order Lemma: $\operatorname{ord}_n(a) \mid \varphi(n)$, and $\operatorname{ord}_p(a) \mid p - 1$.

Proof. Let $r_1, \ldots, r_k$ be the residues coprime to n, where $k = \varphi(n)$. The numbers $ar_1, \ldots, ar_k$ are also coprime to n, and no two are congruent, since $ar_i \equiv ar_j$ gives $r_i \equiv r_j$ after cancelling a. So they are the same k residues in a different order, and multiplying everything together gives $a^k r_1 \cdots r_k \equiv r_1 \cdots r_k$. The product $r_1 \cdots r_k$ is coprime to n, so it cancels. □

For a prime, a second proof explains why the order divides $p - 1$. Multiplication by a shuffles the nonzero residues and splits them into cycles. The cycle through r is $r, ar, a^2r, \ldots$, and it first returns to r after exactly $\operatorname{ord}_p(a)$ steps, since $a^k r \equiv r$ means $a^k \equiv 1$. So every cycle has the same length, and these equal cycles share out the $p - 1$ residues between them.



$x \mapsto 2x \pmod 7$
two cycles of length 3

$x \mapsto 3x \pmod 7$
one cycle of length 6

Orders most often enter olympiad problems through expressions like $a^m + 1$. If an odd prime p divides $a^m + 1$, then $a^{2m} \equiv 1$ but $a^m \equiv -1 \not\equiv 1$, so the order divides $2m$ but does not divide m.

Prime factors of $a^{2^k} + 1$. If an odd prime p divides $a^{2^k} + 1$, then $\operatorname{ord}_p(a) = 2^{k+1}$, and therefore $p \equiv 1 \pmod{2^{k+1}}$.

Proof. The order divides 2^{k+1}, so it is 2^j for some $j \le k + 1$. If $j \le k$ then $a^{2^k} = (a^{2^j})^{2^{k-j}} \equiv 1$, contradicting $a^{2^k} \equiv -1$ (and $1 \not\equiv -1$ because p is odd). So $j = k + 1$, and $2^{k+1} \mid p - 1$. □

So every odd prime factor of $n^2 + 1$ is 1 modulo 4, and every odd prime factor of $n^4 + 1$ is 1 modulo 8.

Example 2. Find all pairs of primes (p, q) such that $p \mid 2^q + 1$ and $q \mid 2^p + 1$.

Solution.
Small cases suggest that $p = q = 3$ is the only answer. To prove it we need something that relates p and q, and the order does exactly that.
Both $2^q + 1$ and $2^p + 1$ are odd, so p and q are odd. From $p \mid 2^q + 1$ we get $2^{2q} \equiv 1 \pmod{p}$

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while $2^q \equiv -1 \not\equiv 1$. Let $d = \mathrm{ord}_p(2)$. Then $d \mid 2q$ and $d \nmid q$. The divisors of $2q$ are $1, 2, q, 2q$, and the condition $d \nmid q$ removes 1 and q. If $d = 2$ then $p \mid 2^2 - 1 = 3$, so $p = 3$. If $d = 2q$ then $2q \mid p - 1$ by Fermat, so $p > q$.
The key observation is that this gives a dichotomy: either $p = 3$ or $p > q$. By symmetry, either $q = 3$ or $q > p$. We cannot have both $p > q$ and $q > p$, so at least one of the primes is 3. If $p = 3$ then $q \mid 2^3 + 1 = 9$, so $q = 3$; if $q = 3$ then likewise $p = 3$. The only pair is $(3, 3)$.

Example 3. Let n be an integer. Prove that every prime factor p of $n^4 - n^2 + 1$ satisfies $p \equiv 1 \pmod{12}$.

Solution.
The expression is not of the form $a^m + 1$, so the lemma above does not apply directly. The key observation is a factorisation: $(n^2 + 1)(n^4 - n^2 + 1) = n^6 + 1$. So $p \mid n^6 + 1$, giving $n^6 \equiv -1$ and $n^{12} \equiv 1 \pmod{p}$.
Some housekeeping. The number $n^4 - n^2 + 1 = n^2(n^2 - 1) + 1$ is odd, so p is odd. Also $p \nmid n$, otherwise $n^4 - n^2 + 1 \equiv 1$. Let $d = \mathrm{ord}_p(n)$. Then $d \mid 12$ and $d \nmid 6$, which leaves $d = 4$ or $d = 12$.
Here is where a careless solution goes wrong: it is tempting to say "so $d = 12$" and stop. Suppose $d = 4$. Then $n^4 \equiv 1$, so $0 \equiv n^4 - n^2 + 1 \equiv 2 - n^2$, giving $n^2 \equiv 2$ and so $n^4 \equiv 4$. Together with $n^4 \equiv 1$ this forces $p \mid 3$. But 3 never divides $n^4 - n^2 + 1$: if $3 \mid n$ the value is 1 modulo 3, and otherwise $n^2 \equiv 1$ and the value is $1 - 1 + 1 = 1$ modulo 3. So $d = 4$ is impossible, $d = 12$, and $12 \mid p - 1$. □

Primitive roots, and why $x^d \equiv 1$ has few solutions

Lagrange's Theorem. Let p be prime and f a polynomial with integer coefficients and degree $d \geq 1$, whose leading coefficient is not divisible by p. Then $f(x) \equiv 0 \pmod{p}$ has at most d solutions modulo p.

Proof. Induct on d; the case $d = 1$ is clear because the leading coefficient is invertible. If r is a root, polynomial division gives $f(x) = (x - r)g(x) + f(r)$ with g of degree $d - 1$, integer coefficients and the same leading coefficient. If $s \not\equiv r$ is another root, then $(s - r)g(s) \equiv 0$, and since p is prime and $p \nmid s - r$, we get $g(s) \equiv 0$. So the other roots are roots of g, and there are at most $d - 1$ of them. □

The word prime matters: $x^2 \equiv 1 \pmod{8}$ has four solutions. A consequence is that k-th powers take few values modulo p (see L4-NT2).

The deeper fact, which we state without proof, is the **Primitive Root Theorem**: for every prime p there is a residue g with $\mathrm{ord}_p(g) = p - 1$. Then $g, g^2, \ldots, g^{p-1}$ run through all the nonzero residues, and for every $d \mid p - 1$ the residue $g^{(p-1)/d}$ has order exactly d. So *an element of order d exists modulo p if and only if $d \mid p - 1$*. The proof counts elements of each order using Lagrange's Theorem; it is a good exercise.

Lifting the exponent

Orders tell you when p divides $a^n - b^n$. The next tool tells you how many times. Write $v_p(m)$ for the exponent of p in m.

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Lifting the Exponent (LTE). Let p be an odd prime and let a, b be integers with $p \mid a-b$ and $p \nmid ab$. Then for every positive integer n,

$$v_p(a^n - b^n) = v_p(a-b) + v_p(n).$$

For $p = 2$: if a, b are odd and n is even, then $v_2(a^n - b^n) = v_2(a-b) + v_2(a+b) + v_2(n) - 1$.

Proof for odd p. Step 1: if $p \nmid n$, write $a^n - b^n = (a-b)S$ with $S = a^{n-1} + a^{n-2}b + \cdots + b^{n-1}$. As $a \equiv b$, each of the n terms of S is congruent to a^{n-1}, so $S \equiv na^{n-1} \not\equiv 0 \pmod p$, and $v_p(a^n - b^n) = v_p(a-b)$.

Step 2: the case $n = p$. Put $a = b + t$ with $p \mid t$. By the binomial theorem $(b+t)^m \equiv b^m + mtb^{m-1} \pmod{t^2}$, and $p^2 \mid t^2$, so modulo p^2

$$S = \sum_{i=0}^{p-1} (b+t)^{p-1-i} b^i \equiv pb^{p-1} + tb^{p-2} \sum_{i=0}^{p-1} (p-1-i) = pb^{p-1} + tb^{p-2} \cdot p \cdot \frac{p-1}{2}.$$

Because p is odd, $\frac{p-1}{2}$ is an integer and the last term is divisible by pt, hence by p^2. So $S \equiv pb^{p-1} \pmod{p^2}$, giving $v_p(S) = 1$ and $v_p(a^p - b^p) = v_p(a-b) + 1$.

For general $n = p^k m$ with $p \nmid m$, apply Step 2 to a, b, then to a^p, b^p, and so on k times (the hypotheses survive each time), and finish with Step 1. □

For $p = 2$, write $n = 2^k m$ with m odd. The odd part changes nothing, by the argument of Step 1 (a sum of m odd terms is odd), and $a^{2^k} - b^{2^k} = (a-b)(a+b)(a^2+b^2)(a^4+b^4)\cdots(a^{2^{k-1}} + b^{2^{k-1}})$, where each factor $a^{2^j} + b^{2^j}$ with $j \geq 1$ is a sum of two odd squares, so it is 2 modulo 4. That gives the extra $k - 1$.

Example 4. Prove that for every positive integer k, the order of 2 modulo 3^k is $2 \cdot 3^{k-1}$.

Solution.
We want to know exactly which n make $3^k \mid 2^n - 1$, and LTE is the natural tool. But LTE needs $3 \mid a - b$, and $3 \nmid 2 - 1$. So we first climb up to the order modulo 3: since $2 \equiv -1$, we have $2^n \equiv 1 \pmod 3$ if and only if n is even.
So if $3^k \mid 2^n - 1$ then $n = 2m$, and $2^n - 1 = 4^m - 1$ where now $3 \mid 4 - 1$. LTE gives $v_3(4^m - 1) = v_3(3) + v_3(m) = 1 + v_3(n)$. Therefore $3^k \mid 2^n - 1$ if and only if n is even and $v_3(n) \geq k - 1$, that is, if and only if $2 \cdot 3^{k-1} \mid n$. The smallest such positive n is $2 \cdot 3^{k-1}$, which is the order. □
So 2 is a primitive root modulo every power of 3. The pattern "find the order modulo p, then lift with LTE" is what AIME questions about the least n with $p^k \mid a^n - b^n$ need.

Common traps.

- The order only exists when $\gcd(a, n) = 1$. Check this before you write $\mathrm{ord}_n(a)$.
- The order divides $p - 1$ but is usually smaller. Never assume 2 has order $p - 1$ without checking.
- From $a^m \equiv -1$ you learn that the order divides $2m$ but not m. List all such divisors, including the small ones: the cases $d = 2$ in Example 2 and $d = 4$ in Example 3 are exactly where marks are lost.
- LTE needs $p \mid a - b$ and $p \nmid ab$. If $p \nmid a - b$, first raise to the order. Using the odd-prime formula at $p = 2$ is wrong: $v_2(3^2 - 1) = 3$, not $v_2(2) + v_2(2) = 2$.
- Lagrange's bound needs a prime modulus.

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How to spot this in a contest

Think orders whenever a problem involves $a^n \pm 1$, $a^n - b^n$ or "for which n does p divide": the answer is usually a congruence class modulo an order. Prime factors of n^2+1, n^4+1 or anything dividing $n^k \pm 1$ point to the "order divides $2m$ but not m" argument. "Largest power of p dividing" or "least n with $p^k \mid \ldots$" is LTE, after first reaching the order modulo p. Exponent towers are reduced by working modulo the order (or φ) in the exponent.

Finally, a standard opening for problems of the form $n \mid a^n - b^n$ or $n \mid a^n + 1$: look at the **smallest prime factor** p of n. The order of a (or of ab^{-1}) modulo p divides both something built from n and $p-1$, and since every prime factor of n is at least p while every prime factor of $p-1$ is less than p, that common divisor is tiny. You will use this in the practice set.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ Find the order of 13 modulo 100, and use it to find the last two digits of 13^{2026}.

2. ⋆⋆ Find all positive integers n such that 13 divides $2^n + 3^n$.

3. ⋆⋆ Find the largest integer k such that 10^k divides $3^{2000} - 1$.

4. ⋆⋆⋆ Find the sum of all primes $p < 200$ for which there is an integer n with $p \mid n^6 + n^3 + 1$. (You may use the Primitive Root Theorem.)

5. ⋆⋆⋆ Prove that there are infinitely many primes of the form $8k+1$.

6. ⋆⋆⋆⋆ Let n be an odd positive integer such that n divides $3^n + 1$. Prove that $n = 1$.

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. UKMT BMO1 2005/06 Problem 1 ⋆
2. AIME 1983 Problem 6 ⋆
3. UKMT BMO1 2006/07 Problem 1 ⋆⋆
4. AIME I 2011 Problem 11 ⋆⋆
5. UKMT BMO1 2013/14 Problem 3 ⋆⋆⋆
6. AIME I 2020 Problem 12 ⋆⋆⋆
7. AIME I 2019 Problem 14 ⋆⋆⋆
8. AIME II 2021 Problem 13 ⋆⋆⋆
9. UKMT BMO1 2024/25 Problem 5 ⋆⋆⋆⋆
10. AIME I 2024 Problem 13 ⋆⋆⋆⋆
11. AIME II 2023 Problem 15 ⋆⋆⋆⋆

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Hints

1. Work modulo 4 and modulo 25 separately; the order modulo 100 is the lowest common multiple of the two orders.
2. Divide through by 3^n: the condition is about a single number raised to the power n.
3. Find the powers of 2 and of 5 separately. For 5, you need to reach the order of 3 modulo 5 before LTE applies.
4. Multiply by $n^3 - 1$. What can the order of n modulo p be?
5. Suppose there are only finitely many and build a number of the form $N^4 + 1$ from them. What do its prime factors look like?
6. Suppose $n > 1$ and let p be the smallest prime factor of n. The order of 3 modulo p divides two numbers with very little in common.

Solutions

Problem 1

Answer: The order is 20; the last two digits are 09.

Modulo 4 we have $13 \equiv 1$, so the order there is 1. Modulo 25 the order divides $\varphi(25) = 20$. Now $13^2 = 169 \equiv 19 \equiv -6$, so $13^4 \equiv 36 \equiv 11$ and $13^5 \equiv 143 \equiv -7$, giving $13^{10} \equiv 49 \equiv -1 \pmod{25}$. Hence $13^{20} \equiv 1$, while $13^{10} \not\equiv 1$ and $13^4 \not\equiv 1$. Every proper divisor of 20 divides 10 or 4, so the order modulo 25 is 20.

A power 13^n is 1 modulo 100 exactly when it is 1 modulo both 4 and 25, that is, when $20 \mid n$. So $\operatorname{ord}_{100}(13) = 20$. Since $2026 = 20 \cdot 101 + 6$, the Order Lemma gives $13^{2026} \equiv 13^6 \pmod{100}$. From $13^2 \equiv 69$ we get $13^4 \equiv 69^2 = 4761 \equiv 61$ and $13^6 \equiv 61 \cdot 69 = 4209 \equiv 9$. The last two digits are 09.

Problem 2

Answer: Exactly the n with $n \equiv 2 \pmod 4$.

Since $3 \cdot 9 = 27 \equiv 1 \pmod{13}$, the inverse of 3 is 9. Multiplying by 9^n, the condition $2^n + 3^n \equiv 0$ becomes $(2 \cdot 9)^n \equiv -1$, that is, $5^n \equiv -1 \pmod{13}$.

Now $5^2 = 25 \equiv -1$, so $5^4 \equiv 1$ and the order of 5 is 4. The powers of 5 cycle through $5, -1, -5, 1$, so $5^n \equiv -1$ exactly when $n \equiv 2 \pmod 4$. These are the required n; for example $2^2 + 3^2 = 13$.

Problem 3

Answer: 4

For the prime 2: 3 and 1 are odd and 2000 is even, so the $p = 2$ form of LTE gives $v_2(3^{2000} - 1) = v_2(3 - 1) + v_2(3 + 1) + v_2(2000) - 1 = 1 + 2 + 4 - 1 = 6$.

For the prime 5: LTE cannot be applied to $3^{2000} - 1^{2000}$ directly, because $5 \nmid 3 - 1$. But $3^2 = 9 \equiv -1 \pmod 5$, so $\operatorname{ord}_5(3) = 4$, and $3^{2000} - 1 = 81^{500} - 1$ with $5 \mid 81 - 1$. LTE now gives $v_5(81^{500} - 1) = v_5(80) + v_5(500) = 1 + 3 = 4$.

So $3^{2000} - 1$ is divisible by 2^6 and by 5^4 but not by 5^5, and the largest power of 10 dividing it is 10^4. Thus $k = 4$.

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Problem 4

Answer: 911

Note first that $n^6 + n^3 + 1$ is always odd, since $n^6 + n^3 = n^3(n^3 + 1)$ is even; so $p \neq 2$. Also $(n^3 - 1)(n^6 + n^3 + 1) = n^9 - 1$.

Suppose $p \mid n^6 + n^3 + 1$. Then $p \nmid n$ and $n^9 \equiv 1 \pmod{p}$, so $d = \operatorname{ord}_p(n)$ divides 9. If $d \mid 3$ then $n^3 \equiv 1$, so $n^6 + n^3 + 1 \equiv 3$ and $p = 3$. Otherwise $d = 9$, and $9 \mid p - 1$ by Fermat's Little Theorem.

Conversely, $p = 3$ works with $n = 1$, since $1 + 1 + 1 = 3$. If $p \equiv 1 \pmod{9}$, the Primitive Root Theorem gives an n of order 9 modulo p. Then $p \mid n^9 - 1 = (n^3 - 1)(n^6 + n^3 + 1)$, while $p \nmid n^3 - 1$ because the order is not a divisor of 3. Hence $p \mid n^6 + n^3 + 1$.

So the primes are 3 together with the primes below 200 that are 1 modulo 9. Running through $10, 19, 28, \ldots, 199$, these are $19, 37, 73, 109, 127, 163, 181, 199$ (the others, such as $91 = 7 \cdot 13$ and $145 = 5 \cdot 29$, are composite or even). The sum is $3 + 19 + 37 + 73 + 109 + 127 + 163 + 181 + 199 = 911$.

Problem 5

Suppose, for contradiction, that there are only finitely many primes congruent to 1 modulo 8; call them $p_1, \ldots, p_r$ (possibly $r = 0$). Let $N = 2p_1p_2 \cdots p_r$, with $N = 2$ if $r = 0$, and let p be any prime factor of $N^4 + 1$.

Since N is even, $N^4 + 1$ is odd, so p is odd. Also $p \nmid N$, since otherwise $p \mid 1$. From $N^4 \equiv -1 \pmod{p}$ we get $N^8 \equiv 1$, so $d = \operatorname{ord}_p(N)$ divides 8. If d divided 4 we would have $N^4 \equiv 1$, hence $1 \equiv -1$ and $p \mid 2$, which is false. Therefore $d = 8$. By Fermat's Little Theorem and the Order Lemma, $8 \mid p - 1$, so $p \equiv 1 \pmod{8}$.

So p is one of $p_1, \ldots, p_r$, and hence $p \mid N$. This contradicts $p \nmid N$. Therefore there are infinitely many primes of the form $8k + 1$. □

Problem 6

Suppose, for contradiction, that $n > 1$, and let p be the smallest prime factor of n. Since n is odd, p is odd. Also $p \neq 3$, because $3^n + 1 \equiv 1 \pmod{3}$ and p divides $3^n + 1$.

From $p \mid 3^n + 1$ we get $3^n \equiv -1$ and so $3^{2n} \equiv 1 \pmod{p}$. Let $d = \operatorname{ord}_p(3)$, which exists as $p \neq 3$. By the Order Lemma $d \mid 2n$, and by Fermat's Little Theorem $d \mid p - 1$. Hence $d \mid \gcd(2n, p - 1)$.

We claim $\gcd(2n, p - 1) = 2$. Let q be an odd prime dividing this gcd. Then $q \mid n$, so $q \geq p$ by the choice of p; but also $q \mid p - 1$, so $q \leq p - 1$. This is impossible, so the gcd is a power of 2. As n is odd, $4 \nmid 2n$, so the gcd divides 2; and it equals 2 since $p - 1$ is even.

Therefore $d \mid 2$, so $3^2 \equiv 1 \pmod{p}$, that is, $p \mid 8$. This contradicts p being odd. Hence $n = 1$, which does satisfy $1 \mid 3 + 1$. □

Remark. The oddness of n is essential: $n = 2$ and $n = 10$ both divide $3^n + 1$, since $3^{10} + 1 = 59050$.

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